PROJECT AI for COMMUNICATION and MARKETING

Develop Churn Model

1. Business context and project objectives
2. Dataset description and key variables
3. Analytical methodology used in model development
4. Assessment of the business insights generated and practical recommendations

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1 Business context and project objectives

BUSINESS CONTEXT

The Chief Marketing Officer of a major e-commerce retailer notes that an increasing number of customers are abandoning the site, significantly impacting order value and customer lifetime value.

PROJECT OBJECTIVES

To counter this trend, the Data & AI team has been tasked with developing a predictive churn-model capable of identifying customers at high-risk of churning.

The Data & AI team then proceeded to analyse and test the various models:

- Decision Tree,
- Random Forest,
- Logistic Regression,
- Gradient Boosting (XGB and Cat Boost),
- Neural Network (Multi-Layer Perceptron)

and considered the most suitable and effective models for the research.

The goals are:

- to understand where the risk of customer churn lies, enabling the company to plan and predict targeted marketing and loyalty campaigns.
- to grow the customer retention, improving the marketing effectiveness and sustaining revenue growth.

2 Dataset description and key variables

DATASET DESCRIPTION

The dataset consists of 5630 records and 20 columns (attributes).
For each customer, we have different variables describing behavior and demographics. The features are of mixed types: integers, floats, and categorical (object). Several columns contain missing values and outliers, which may require appropriate preprocessing.
The dataset provides a rich base for customer segmentation, churn prediction, and behavioral analytics.

KEY VARIABLES

To identify which variables have the greatest positive or negative influence on churn, in the exploration analysis section it has been used a correlation matrix with respect to numerical variables (int, float). The main significant variables with impact on churn are:

- **Tenure** (correlation = -0.34): customers with longer tenure are less likely to abandon the service.
- **Complain** (correlation = 0.25): customers who have made complaints are more likely to churn.
- **CashbackAmount** (correlation = -0.15): a higher cashback is associated with a lower probability of churn.

These relationships identified by the correlation matrix suggest that customer loyalty, complaint management and financial incentives can play a key role in reducing churn.

3.1 Analytical methodology used in model development

DATA PREPARATION and EXPLORATION

PRE-PROCESSING

- Analysis of the dataset structure
- Analysis of missing values. Missing values in the various columns (Tenure, WarehouseToHome, etc.) were checked using visualisation tools such as missingno.

DATA CLEANING

- Removal of customer id: CustomerID is a unique identifier that does not provide useful information for predicting churn. Since the model to be developed will be applied to known customers, this feature must be excluded to avoid overfitting and improve generalisation.
- Null values: Various methods were used to manage null values, selected with the aim of maintaining the same distribution before and after the cleaning operation .

DATA TYPE CONVERSION

- Identification of decimal values in numerical columns, with the exception of “CashbackAmount” for which decimal numbers are allowed, and conversion to integer numbers.
- Variables (CouponUsed, OrderCount) that have decimal values are rounded and converted to integer numbers to ensure consistency in data management.

OUTLIERS

- Identification and removal of OUTLIERS present in certain numerical variables (NumberOfAddress, DaySinceLastOrder, CashbackAmount).
- The intervention required the elimination of a small number of records, approximately 10 in total, taking care to preserve the structure of the dataset and with the goal of improving model performance.

EXPLORATIVE DATA ANALYSIS (EDA)

- Visualisation of the distribution of numerical and categorical variables.
- In addition, the correlation matrix was used to identify the variables that have the greatest impact and influence on churn .

3.2 Analytical methodology used in model development

CHURN MODEL

ONE-HOT ENCODING

- One-hot encoding was performed in order to enable the use of SMOTE, which requires all variables to have numerical values. At the beginning of the process, categorical variables were kept as ‘strings’ to facilitate analysis and distribution (on EDA).

DATA PARTITIONING

- The dataset was divided into TRAINING and TEST using an 80/20 ratio.

DATA BALANCING

- The use of SMOTE at X train was necessary to solve the problem of the ‘imbalance’ of the target variable Churn, thus ensuring that the model did not become biased towards the ‘dominant’ class.

DATA STANDARDIZATION (only to Logistic Regression and Multi-Layer Perceptron)

- Data standardization was applied only to the Logistic Regression and Multi-Layer Perceptron models using Standard Scaler, as it is important for ensuring proper convergence and optimal performance in these cases. For the other models, scaling was not performed since it does not significantly affect their performance.

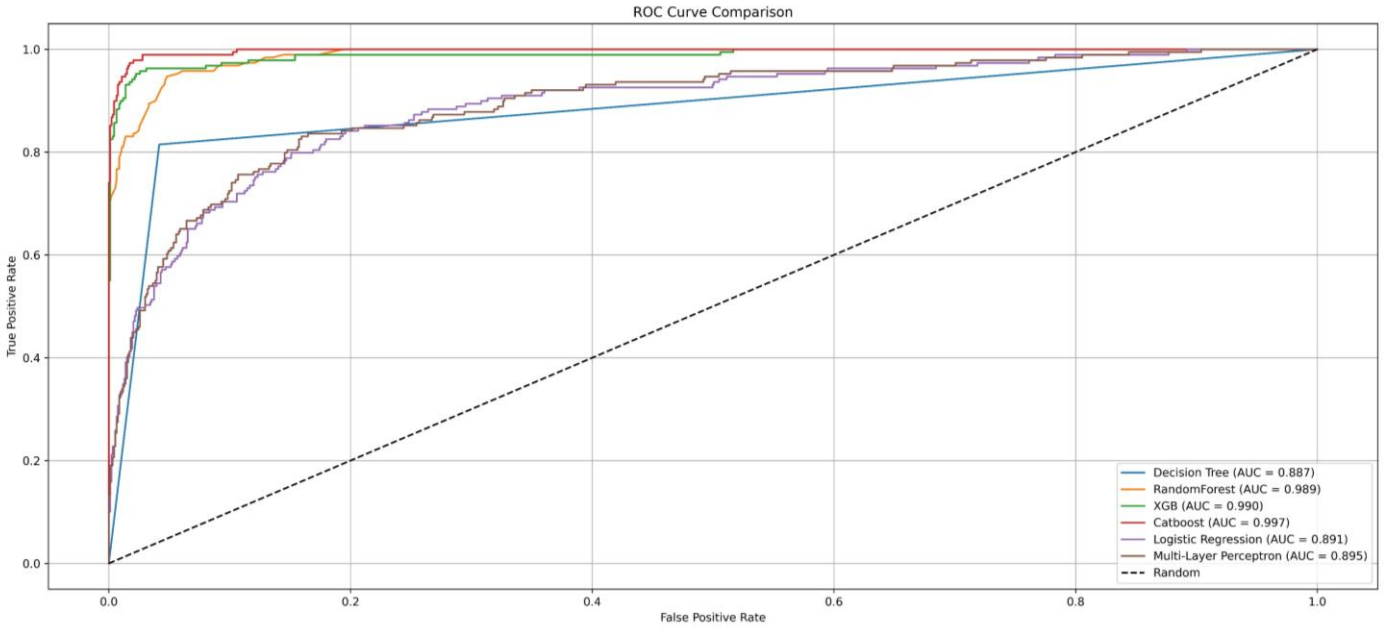
MODELS

- Several models were used and optimised, such as Decision Tree, Random Forest, Logistic Regression, Gradient Boosting (XGB and Cat Boost) and Neural Network (Multi-Layer Perceptron), in order to verify which of these performed best.
- For each model, the following were studied :
  - The ROC curve,
  - Performance scores
  - The features that have the greatest impact on the model
  - The confusion matrix.

3.3 Analytical methodology used in model development

FINAL EVALUATION

The final evaluation of the model's performance took into account the ROC curves and scores of all models explored, and the result obtained is the following:



	roc_auc	recall	precision	accuracy
Decision Tree	0.89	0.81	0.8	0.93
Random Forest	0.99	0.83	0.92	0.96
XGBoost	0.99	0.88	0.95	0.97
CatBoost	1.0	0.93	0.96	0.98
Logistic Regression	0.89	0.65	0.67	0.89
Multi-Layer Perceptron	0.78	0.62	0.69	0.89

By comparing the ROC curves and scores, it was found that **XGBoost** and **CatBoost** are the best-performing models, with CatBoost performing slightly better than XGBoost for most key metrics .

4.1 Assessment of the business insights generated and practical recommendations

DATA-DRIVEN BUSINESS INSIGHTS – the selected models detect:

KEY DRIVERS (Common across all models)

- **Tenure (Customer Longevity)**: It is the most crucial factor; customers with a shorter connection to the company are significantly more likely to leave. This shows how loyalty is built over time and the first few months are the period most at risk.
- **MaritalStatus**: Marital status emerges as a significant element in all models. This suggests that different life stages and contexts (e.g. single, married, divorced) have a profound influence on purchasing habits and brand loyalty.

RELEVANT FACTORS (Important in multiple models)

- **WarehouseToHome (Distance from Warehouse to Customer Home)**: Logistics has a direct impact. Longer distances translate into longer delivery times or higher costs, generating dissatisfaction and increasing the risk of churn.
- **CashbackAmount (Monetary Incentives)**: Customers are sensitive to monetary rewards. The amount of cashback is an indicator of the level of loyalty and the perception of the value offered by the platform.
- **DaySinceLastOrder (Purchase Recency)**: A customer who has not purchased for several days can be a sign of disinterest and a predictor of abandonment.

NICHE INDICATORS

- **Customer Experience**: Complain and SatisfactionScore are decisive factors. A single complain or a low satisfaction score are explicit warning signals.
- **Purchase Behavior**: PreferedOrderCat (preferred category) and OrderAmountHikeFromYear (spending growth) indicate what a customer buys and whether he is increasing his spending and, consequently, his value to the company; they are important predictors.
- **Demographic Profile and Payment Preferences**: Gender, NumberOfAddress and PreferredPaymentMode show that demographic profile and operational payment preferences can also provide important information for risk segmentation.

The 5 Most Influential Features per Model

DECISION TREE	Tenure	MaritalStatus	WarehouseToHome	CashbackAmount	NumberOfAddress
RANDOM FOREST	Tenure	MaritalStatus	CashbackAmount	DaySinceLastOrder	WarehouseToHome
XGBOOST	Tenure	MaritalStatus	PreferedOrderCat	PreferredPaymentMode	Gender
CATBOOST	Tenure	MaritalStatus	NumberOfAddress	SatisfactionScore	Complain
LOGISTIC REGRESSION	PreferredPaymentMode	MaritalStatus	PreferredLoginDevice	PreferedOrderCat	Gender
MULTI-LAYER PERCEPTRON	MaritalStatus	PreferredPaymentMode	Tenure	PreferredLoginDevice	PreferedOrderCat

4.2 Assessment of the business insights generated and practical recommendations

STRATEGIC RECOMMENDATIONS - Based on these insights, the following strategic actions to reduce churn are recommended:

IMPLEMENT A NEW CUSTOMER ONBOARDING PROGRAM (Based on Tenure)

- **What to do**: Create a 5-10 day “welcome” path for new customers. This path should include targeted communications, tutorials on how to use the platform and exclusive offers to incentivize second and third purchases, consolidating the relationship right from the start.

DEVELOP LIFE-STAGE-BASED MARKETING CAMPAIGNS (Based on MaritalStatus)

- **What to do**: Do not communicate in a generic way but create clusters (behavioural segmentation) of customers based on marital status (and gender, as suggested by XGBoost) to customize offers. For example, promotions on home or family products for married customers or offers on categories such as travel or technology for single customers.

CREATE A PROACTIVE CHURN ALERT SYSTEM (Based on Complain, SatisfactionScore, DaySinceLastOrder)

- **What to do**: Set up automatic triggers. When a customer gives a negative review (low SatisfactionScore), makes a complaint (Complain = 1) or has not purchased for more than X days (DaySinceLastOrder), the system must notify the Customer to Success team . This team should have a “recovery package” of to offers available for this kind of e customers (e.g. a discount voucher) by making a courtesy call to solve the problem and prevent abandonment.

OPTIMIZE LOGISTICS AND EXPECTATION MANAGEMENT (Based on WarehouseToHome)

- **What to do**: For customers geographically distant from warehouses, communicate delivery times transparently. Notify the customer of any delivery delays and consider to offering premium shipping options and/or small giveaways to compensate for any extended waits.

REDESIGN THE LOYALTY PROGRAM AROUND CUSTOMER VALUE (Based on CashbackAmount, OrderAmountHikeFromLastYear)

- **What to do**: Structure a dynamic loyalty programme with a “basic” and a “premium” profile, which is acquired by reaching a certain spending threshold. Customers who increase their spending (positive OrderAmountHikeFromYear) and become “premium” would receive higher benefits.